STPM 2014 PHYSICS 960/3

Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ J s, $c = 3.00 \times 10^{8}$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, $m_p = 1.67 \times 10^{-27}$ kg, $N_A = 6.02 \times 10^{23}$ mol^{-1}, $g = 9.81$ m s^{-2}.

Section A [15 marks]

1. An object of mass m undergoes a simple harmonic motion with frequency f and amplitude A. The maximum restoring force is

A fmA B f^2mA C $2\pi fmA$ D $4\pi^2f^2mA$

Answer: D

$$F_{max} = ma_{max} = m\omega^2 A = 4\pi^2 f^2 mA$$

Explanation: The restoring force is greatest at maximum displacement.

2. Which statement is true of a critically damped oscillation?

A The system undergoes a few oscillations before it stops.
B The system returns to rest at the equilibrium position slowly.
C The system undergoes exponential drop in amplitude when oscillating.
D The system returns to rest at the equilibrium position in the shortest time.

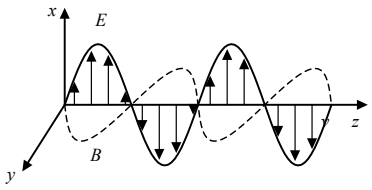
Answer: D

A critically damped system returns to rest at the equilibrium position in the shortest possible time, without oscillating.

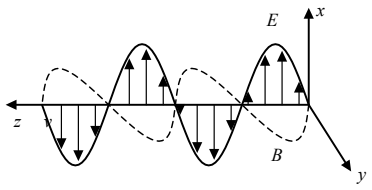
Explanation: A describes underdamping; B describes overdamping.

3. Which graph shows the planes of oscillation of electric field E and magnetic field B of an electromagnetic wave and its velocity v correctly?

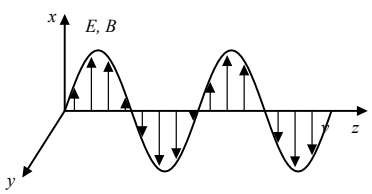
A



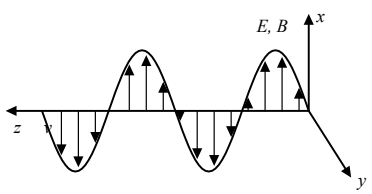
B



C



D

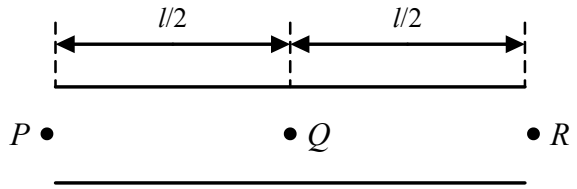


Answer: A

E and **B** are perpendicular to each other and to the velocity, which is along **E** × **B**: **E** along x, **B** along y, **v** along $+z$ (A).

Explanation: E and B oscillate in phase in perpendicular planes.

4. A pipe of length l opens at both ends is shown in the diagram below.



If a stationary sound wave at its fundamental mode is set up in the pipe, which statement is **not** true of the motion of particles P, Q and R?

A Particle Q is always stationary.
B Particle P and R are oscillating in phase.
C Particle Q experiences maximum pressure variation.
D Particle P and R are oscillating with maximum amplitude.

Answer: B

In the fundamental mode of an open pipe, P and R are displacement antinodes on either side of the node Q, so they oscillate in antiphase, not in phase. Statement B is not true.

Explanation: Points on opposite sides of a node in a standing wave move in opposite directions.

5. An observer moves with constant speed toward a stationary source producing sound of frequency f. Which combination of the apparent frequency and the intensity of the sound observed is correct?

	Apparent frequency	*Intensity*
A	Lower than f	Remains unchanged
B	Equals to f	Increases
C	Higher than f	Remains unchanged
D	Higher than f	Increases

Answer: D

Moving towards the source: more wavefronts are met per second (frequency higher than f), and as the observer gets closer the intensity increases.

Explanation: For a point source the intensity $\propto 1/r^2$.

6. Which phenomenon is due to superposition of sound waves?

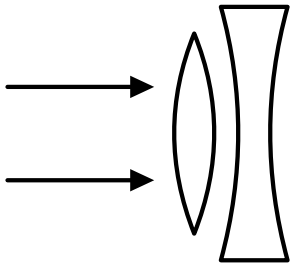
A Beats
B Refraction
C Reflection
D The Doppler effect

Answer: A

Beats are produced by the superposition of two sound waves of slightly different frequencies.

Explanation: Refraction, reflection and the Doppler effect do not involve superposition.

7. A beam of parallel rays incident normally on the converging lens of focal length f_1 placed in contact with a diverging lens of focal length f_2 is shown in the diagram below.



If the focal length of the lenses combination is f_T, and the distance of the image formed by the converging lens and diverging lens are v_1 and v_2 respectively, which is true of v_1 and v_2?

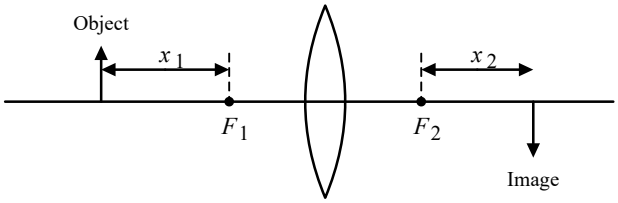
	v_1	v_2
A	f_1	f_2
B	f_1	f_T
C	f_2	f_1
D	f_2	f_T

Answer: B

The converging lens alone would form an image at its focal point: $v_1 = f_1$. The combination focuses parallel light at its combined focal length: $v_2 = f_T$.

Explanation: The image from the first lens acts as a (virtual) object for the second lens.

8. A concave lens with focal length f formed an image of the object is shown in the diagram below.



F_1 and F_2 are the focus points of the lens. x_1 is the distance of the object from F_1 and x_2 is the distance of the image from F_2. The distances x_1 and x_2 can be expressed in terms of f as

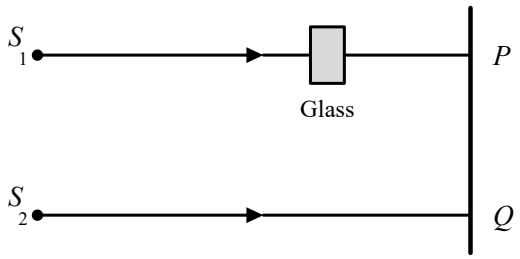
A $x_1x_2 = f^2$ B $x_1^2 + x_2^2 = f^2$ C $\frac{x_1x_2}{x_1 + x_2} = f$ D $\frac{x_1x_2}{x_1 - x_2} = f$

Answer: A

Newton's lens formula: $x_1x_2 = f^2$

Explanation: It follows from $1/u + 1/v = 1/f$ with $u = f + x_1$ and $v = f + x_2$.

9. Two parallel light rays from two sources S_1 and S_2 are shown in the diagram below. The geometrical distances S_1P and S_2Q are equal.



If the ray from S_1 passes through a piece of glass of thickness 1.0 mm and refractive index 1.5, what is the optical path difference between the two rays?

A 0 B 0.3 mm C 0.5 mm D 1.0 mm

Answer: C

Optical path difference = $(n - 1)t = (1.5 - 1)(1.0 \text{ mm}) = 0.5 \text{ mm}$

Explanation: The glass replaces 1.0 mm of air with an optical path of 1.5 mm.

10. Silicon films become better electric conductors when illuminated with photons of energies greater than 1.14 eV. What is the corresponding maximum wavelength of the photon?

A 3.62×10^{-15} m B 4.71×10^{-15} m C 5.47×10^{-11} m D 1.09×10^{-6} m

Answer: D

$$\lambda_{max} = \frac{hc}{E} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{(1.14)(1.60 \times 10^{-19})} = 1.09 \times 10^{-6} \text{ m}$$

Explanation: Lower photon energy means longer wavelength, so the minimum energy corresponds to the maximum wavelength.

11. In a photoelectric effect experiment, monochromatic light of wavelength 400 nm is irradiated on a metal surface to produce photoelectrons. If the stopping potential is 1.20 V, what is the threshold wavelength of the metal?

A 289 nm B 400 nm C 652 nm D 1040 nm

Answer: C

$$W = \frac{hc}{\lambda} - eV_s = 4.97 \times 10^{-19} - 1.92 \times 10^{-19} = 3.05 \times 10^{-19}\ \text{J}$$

$$\lambda_0 = \frac{hc}{W} = 6.52 \times 10^{-7}\ \text{m} = 652\ \text{nm}$$

Explanation: The threshold wavelength is longer than 400 nm because 400 nm light ejects electrons.

12. In an X-ray tube, the continuous spectrum is produced when

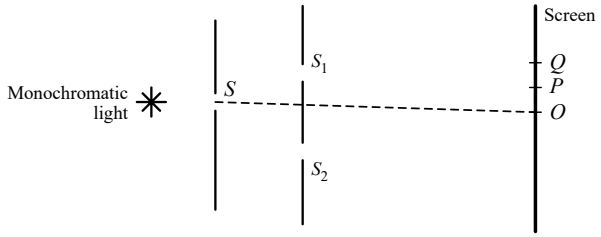
A the mass of an electron is converted into a photon during deceleration
B an electron in the innermost orbit is knocked out from the target atom
C the kinetic energy of an electron is radiated as photon during deceleration
D an electron in the target atom makes a transition from outer orbit to the innermost orbit

Answer: C

The continuous spectrum comes from electrons losing kinetic energy as they decelerate in the target; the energy lost is radiated as X-ray photons (bremsstrahlung).

Explanation: Knocking out inner electrons and the transitions that follow produce the characteristic lines.

13. The arrangement of Young two-slit experiment is shown in the diagram below.



Points O, P and Q at the screen are centre maximum, first minimum and first maximum respectively. What is the phase difference between waves from S_1 and S_2 when they reach O, P and Q?

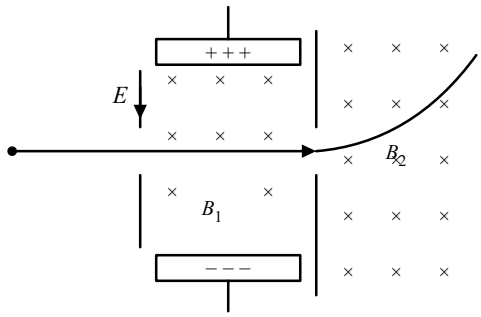
	O	P	Q
A	0	$\frac{\pi}{2}$	π
B	0	π	2π
C	0	$\frac{3}{2}\pi$	3π
D	0	2π	4π

Answer: B

Path differences 0, ½λ and λ correspond to phase differences 0, π and 2π.

Explanation: Phase difference = $(2\pi/\lambda)$ × path difference.

14. A positive ion of charge q and mass m entering a velocity selector of electric field E and magnetic field B_1 is shown in the diagram below.



If the ion then enters a region of magnetic field B_2 and moves in a circular path, what is the radius of the path?

A $\frac{mE}{B_1^2 q}$ B $\frac{mE}{B_2^2 q}$ C $\frac{mE}{B_1 B_2 q}$ D $\frac{mEq}{B_1 B_2}$

Answer: C

$$\text{Velocity selector: } qE = qvB_1 \Longrightarrow v = \frac{E}{B_1}$$

$$\text{In } B_2\text{: } r = \frac{mv}{qB_2} = \frac{mE}{B_1 B_2 q}$$

Explanation: This is the principle of the Bainbridge mass spectrometer.

15. A radioactive sample of a certain nuclide with a molar mass of 32 g mol^{-1} produces an activity of 4.0×10^7 s^{-1}. If the decay constant is 5.6×10^{-7} s^{-1}, what is the mass of the sample?

A 1.2×10^{-10} g B 3.8×10^{-9} g C 7.1×10^{13} g D 2.3×10^{15} g

Answer: B

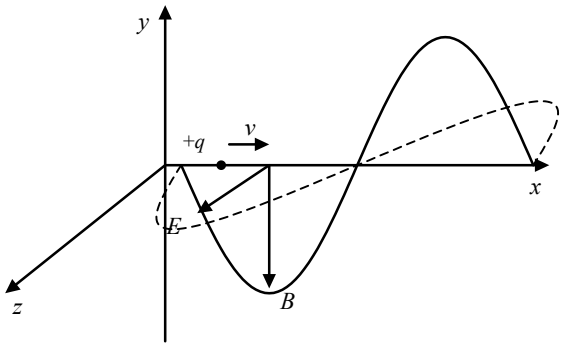
$$N = \frac{A}{\lambda} = \frac{4.0 \times 10^7}{5.6 \times 10^{-7}} = 7.1 \times 10^{13}$$

$$\text{Mass} = \frac{7.1 \times 10^{13}}{6.02 \times 10^{23}} \times 32 \text{ g} = 3.8 \times 10^{-9} \text{ g}$$

Explanation: Number of moles = number of nuclei ÷ Avogadro constant.

Section B [15 marks]

16. (a) A positive charge moves in the positive-x direction of an electromagnetic wave as shown in the diagram below.



The charge moves with a constant velocity of 2.0×10^5 m s^{-1}.

(i) Why is the charge likely to keep moving with the constant velocity? [2]

Answer

The charge experiences an electric force qE and a magnetic force qvB in opposite directions 1

They are equal in magnitude, so the resultant force is zero and the velocity stays constant 1

2

Explanation: This is the condition used in a velocity selector.

(ii) If the amplitude of the electric field is 8.0×10^4 N C^{-1}, determine the amplitude of the magnetic field. [2]

Answer

$qE = qvB \Longrightarrow B = \frac{E}{v} = \frac{8.0 \times 10^4}{2.0 \times 10^5}$ 1

$B = 0.40$ T 1

2

Explanation: Use the speed of the charge, not the speed of light.

(b) An electromagnetic wave propagates in a material with a speed of 2.10×10^8 m s^{-1} and a frequency of 1.20×10^{10} Hz. Determine the

(i) wavelength in air, [2]

Answer

$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8}{1.20 \times 10^{10}}$ 1

$\lambda = 2.50 \times 10^{-2}$ m 1

2

Explanation: The frequency is the same in the material and in air.

(ii) refractive index of the material. [2]

Answer

$n = \frac{c}{v} = \frac{3.00 \times 10^8}{2.10 \times 10^8}$ 1

$n = 1.43$ 1

2

Total for question **8**

Explanation: Refractive index = speed in vacuum ÷ speed in the material.

17. (a) Explain de Broglie equation. [2]

Answer

A moving particle has wave properties; its de Broglie wavelength is $\lambda = \frac{h}{p} = \frac{h}{mv}$ 1

The wavelength is inversely proportional to the momentum of the particle 1

2

Explanation: This wave–particle duality is confirmed by electron diffraction.

(b)

(i) Calculate the ratio of the kinetic energy of electron and the kinetic energy of proton if both of them have the same de Broglie wavelength. [3]

Answer

Same wavelength $\Longrightarrow$ same momentum p — 1

$K = \frac{p^2}{2m} \Longrightarrow \frac{K_e}{K_p} = \frac{m_p}{m_e}$ — 1

$= \frac{1.67 \times 10^{-27}}{9.11 \times 10^{-31}} = 1.83 \times 10^3$ — 1

3

Explanation: The much lighter electron needs far more kinetic energy for the same momentum.

(ii) If the proton has de Broglie wavelength of 6.0×10^{-13} m, calculate the potential needed to accelerate it from rest to achieve this wavelength. [2]

Answer

$eV = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2} \Longrightarrow V = \frac{(6.63 \times 10^{-34})^2}{2(1.67 \times 10^{-27})(1.60 \times 10^{-19})(6.0 \times 10^{-13})^2}$ — 1

$V = 2.3 \times 10^3$ V — 1

2

Total for question **7**

Explanation: The kinetic energy gained is the charge times the accelerating potential.

Section C [30 marks]

18. (a) Define *simple harmonic motion*. [2]

Answer

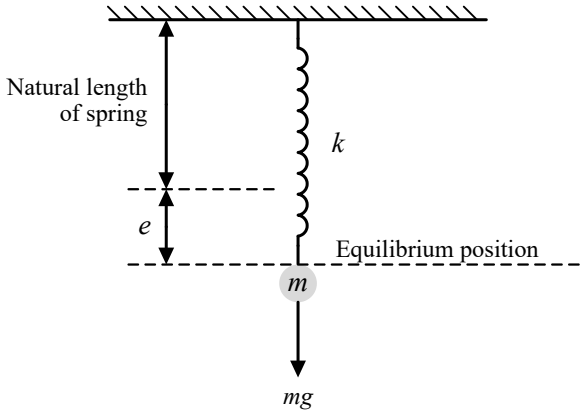
The acceleration is directly proportional to the displacement from a fixed point (the equilibrium position) — 1

and always directed towards that point: $a = -\omega^2 x$ — 1

2

Explanation: Both parts are required.

(b) A mass m hangs vertically on a spring with spring constant k, as shown in the diagram below.



The extension e is 1.60 cm. The mass is pulled down a small distance less than e and released to oscillate around the equilibrium position. At time t, the displacement of the mass from equilibrium position is x.

(i) Show that the mass oscillates as a simple harmonic motion. [5]

Answer

At equilibrium: $ke = mg$	1
At displacement x below equilibrium: resultant force $= mg - k(e + x)$	1
$= -kx$ (upward, towards equilibrium)	1
$ma = -kx \Longrightarrow a = -\frac{k}{m}x$	1
k/m is constant: SHM with $\omega^2 = \frac{k}{m} = \frac{g}{e}$	1
	5

Explanation: The weight is cancelled by the tension at equilibrium, so only the extra spring force acts as the restoring force.

(ii) Determine the period for the oscillation. [2]

Answer

$T = 2\pi\sqrt{(\frac{e}{g})} = 2\pi\sqrt{(\frac{0.0160}{9.81})}$	1
$T = 0.254$ s	1
	2

Explanation: Since $k/m = g/e$, the period depends only on the static extension.

(c) A mass of 60.0 g performs a simple harmonic motion represented by $y = 5.0 \sin 6.2t$, where y is the displacement in centimetre and t is the time in second. Determine the

(i) maximum kinetic energy of the system, [3]

Answer

$\omega = 6.2$ rad s^{-1}, $A = 5.0$ cm $= 0.050$ m — 1

$K_{max} = \frac{1}{2}m\omega^2A^2 = \frac{1}{2}(0.0600)(6.2)^2(0.050)^2$ — 1

$K_{max} = 2.88 \times 10^{-3}$ J — 1

3

Explanation: This equals the total energy of the system.

(ii) potential energy of the system at time 0.40 s. [3]

Answer

$y = 5.0 \sin (6.2 \times 0.40) = 5.0 \sin 2.48$ rad $= 3.07$ cm — 1

$U = \frac{1}{2}m\omega^2y^2 = \frac{1}{2}(0.0600)(6.2)^2(0.0307)^2$ — 1

$U = 1.09 \times 10^{-3}$ J — 1

3

Total for question **15**

Explanation: Use radians to find the displacement.

19. (a) A concave mirror has radius of curvature 40.0 cm. A nail of 5.00 cm long is placed upright on the principal axis, 45.0 cm in front of the mirror.

(i) Calculate the distance of the image. [3]

Answer

$f = \frac{40.0}{2} = 20.0$ cm — 1

$\frac{1}{v} = \frac{1}{20.0} - \frac{1}{45.0}$ — 1

$v = 36.0$ cm (in front of the mirror) — 1

3

Explanation: The object is beyond C (40 cm), so the image is between F and C.

(ii) Determine the size of the image. [2]

Answer

$m = \frac{36.0}{45.0} = 0.800$ 1

Image size = (0.800)(5.00) = 4.00 cm 1

2

Explanation: The image is smaller than the nail.

(iii) State **two** characteristics of the image. [2]

Answer

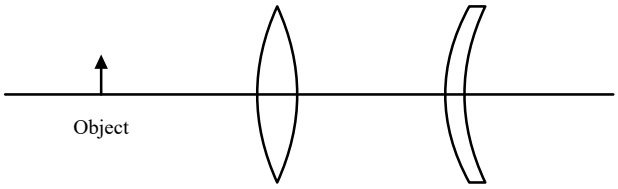
Real and inverted 1

Diminished 1

2

Explanation: Typical for an object beyond the centre of curvature.

(b) An object, a bi-convex lens and a meniscus lens are arranged as shown in the diagram below.



The meniscus lens has radii of curvature 20.0 cm and 15.0 cm respectively, is made of glass with refractive index $n = 1.5$. It is placed 50.0 cm behind the bi-convex lens of focal length 12.0 cm. The object is placed 25.0 cm in front of the bi-convex lens. Determine

(i) the focal length of the meniscus lens, [2]

Answer

$\frac{1}{f} = (1.5 - 1)(\frac{1}{15.0} - \frac{1}{20.0})$ 1

$f = +120$ cm (a converging meniscus) 1

2

Explanation: The surfaces curve the same way, so their effects partly cancel; the more strongly curved surface (15.0 cm) dominates.

(ii) the position of the final image formed by the arrangement, [4]

Answer

Bi-convex lens: $\frac{1}{v_1} = \frac{1}{12.0} - \frac{1}{25.0} \Rightarrow v_1 = 23.1$ cm — 1

Object for the meniscus lens: $u_2 = 50.0 - 23.1 = 26.9$ cm — 1

$\frac{1}{v_2} = \frac{1}{120} - \frac{1}{26.9}$ — 1

$v_2 = -34.7$ cm: a virtual image 34.7 cm in front of the meniscus lens — 1

4

Explanation: The intermediate image lies inside the focal length of the meniscus lens, which then acts as a magnifier.

(iii) the magnification of the final image. [2]

Answer

$m = m_1 m_2 = (-\frac{23.1}{25.0})(\frac{34.7}{26.9})$ — 1

$m = -1.19$ (inverted, 1.19 times the object size) — 1

2

Total for question **15**

Explanation: The first lens inverts the image and the second keeps it the same way up.

20. (a)

(i) Explain the difference between soft X-rays and hard X-rays. [2]

Answer

Soft X-rays: longer wavelength, lower photon energy, low penetrating power (low tube voltage) — 1

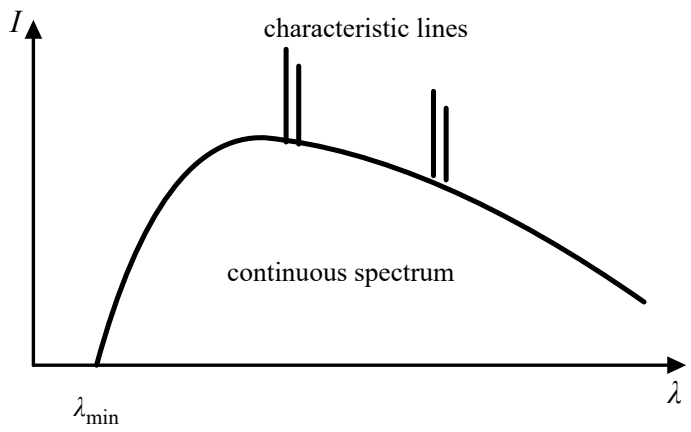
Hard X-rays: shorter wavelength, higher photon energy, high penetrating power (high tube voltage) — 1

2

Explanation: Hardness is controlled by the accelerating voltage.

(ii) Sketch a graph of intensity against wavelength to represent a typical X-ray spectrum. [3]

Answer



Axes labelled (intensity against wavelength)	1
Continuous spectrum with a sharp cut-off λ_{min}	1
Characteristic lines superimposed	1
	3

Explanation: The continuous curve rises from λ_{min} to a maximum and then falls gradually.

(iii) State **three** characteristics of the spectrum. [3]

Answer

• A continuous spectrum with a sharp minimum (cut-off) wavelength	1
• Sharp characteristic lines whose wavelengths depend on the target material	1
• The minimum wavelength depends only on the accelerating voltage ($\lambda_{min} = hc/eV$)	1
	3

Explanation: Increasing the tube current increases the intensity but not the wavelengths.

(b) The potential difference across an X-ray tube is 20 kV and the current flow through the tube is 18 mA. Heat is generated at the rate of 258 W at the anode of the tube.

(i) State a physical property of the substance which is used as the anode. [1]

Answer

A high melting point (e.g. tungsten)	1
	1

Explanation: Most of the electrons' energy becomes heat in the anode.

(ii) Calculate the speed attained by the electron just before hitting the anode. [2]

Answer

$\frac{1}{2}mv^2 = eV \Longrightarrow v = \sqrt{(\frac{2(1.60 \times 10^{-19})(20 \times 10^3)}{9.11 \times 10^{-31}})}$ 1

$v = 8.38 \times 10^7$ m s^{-1} 1

2

Explanation: Non-relativistic calculation.

(iii) Calculate the rate of change of electric energy to X-ray energy of the tube. Explain why only a small percentage of electric energy is turned into X-ray energy. [4]

Answer

Electrical power supplied = $VI = (20 \times 10^3)(18 \times 10^{-3}) = 360$ W 1

Rate of conversion to X-rays = 360 − 258 = 102 W 1

Most electrons lose their energy gradually in many collisions with atoms of the anode, which increases the vibrations of the atoms (heat) 1

Only a very small fraction of the interactions produce X-ray photons 1

4

Total for question **15**

Explanation: With the given data the efficiency is 102/360 = 28%; in practice it is usually only about 1%.